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Advancing the Understanding of Coupled Physical Processes in Porous Media for Multidisciplinary Applications

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Abstract

Understanding of the porous media are critical to gas exploitation. The primary objective of this study is to investigate the relationship between pore size, pore throat characterization and fluid flow using micro-CT Scan and core samples using wave propagation, electron transport, and electrolyte flow within porous structures, emphasizing their elastic, and plastic properties. Understanding these coupled processes is essential for optimizing gas extraction and productivity.

The methodology combines experimental and modeling approaches to analyze the complex interactions within porous media. Advanced imaging and x-ray, elastic velocities, and triaxial testing, were used to characterize pore structural connectivity, heterogeneity and anisotropy. Modeling efforts incorporated fluid flow models, and transport equations to simulate coupled processes and predict porous system behavior. Finite Element Modeling (FEM) and multi-physics simulations were employed to account for the interdependence of geomechanical, and transport mechanisms. These methods enable a detailed understanding of how physical properties like permeability, porosity, and conductivity evolve under various conditions.

The results highlight significant advancements in understanding the interconnectivity of the fluid transport and mechanical properties in porous media. Elastic wave propagation was found to be highly sensitive to anisotropic pore structural variations, offering insights into stress and strain distribution and deformation mechanisms. Fluid transport exhibited nonlinear behavior in heterogeneous systems, influenced by pore geometry and fluid distribution. Electrochemical processes, such as ion migration and charge transport, were shown to interact dynamically with mechanical stress, leading to complex feedback mechanisms. These findings improve the predictability of porous media behavior in different gas fields, such as subsurface fluid flow during gas extraction or storage operations.

In conclusion, this research provides a comprehensive framework for studying coupled physical processes in porous media, bridging gaps in knowledge across multiple scientific domains. The integration of experimental and computational methodologies offers a robust platform for addressing challenges such as heterogeneity, anisotropy, and coupled fluid transport mechanisms.

The innovation lies in the unified approach to modeling and characterizing multi-physical interactions, which has far-reaching implications for improving the efficiency and sustainability of technologies in energy, construction, and environmental sectors. The outcomes of this study contribute to a deeper understanding of fundamental processes governing porous media behavior and provide actionable insights for optimizing industrial applications.

Introduction

Efficient and sustainable exploitation of subsurface energy resources—as well as the safe, long-term storage of gases such as hydrogen (H_2) and carbon dioxide (CO_2)—requires a detailed understanding of the multiphysics interactions within porous materials. These geological systems are subjected to cyclic loading, thermal stimulation, fluid injection and withdrawal, and chemical alterations that drive time-dependent changes in porosity, permeability, and mechanical integrity.

Porous media such as shales, carbonates, coals, and sandstones respond to complex Thermo-Hydro-Mechanical-Chemical (THMC) couplings, which control the transport, deformation, and fracture behavior at multiple scales. Applications including underground gas storage, enhanced oil recovery (EOR) by water or CO_2 injection, gas hydrate dissociation, geothermal energy recovery, and Steam-Assisted Gravity Drainage (SAGD) all demand models that accurately resolve:

- Coupling between fluid flow and mechanical deformation under cyclic injection and withdrawal conditions.
- Thermal effects on stress redistribution, material softening, and pore-pressure diffusion.
- Multiphase fluid dynamics, including gas, oil, and water migration and displacement.
- Time-dependent pore structure evolution through damage, swelling, and chemical reactions.

Challenges Across Key Applications

- Gas Storage: Cyclic injection/withdrawal leads to stress path reversal, pore collapse, and progressive permeability loss. These changes can alter gas deliverability and threaten storage integrity (Espinoza & Santamarina, 2016, hamid et al 2023).
- Enhanced Oil Recovery (EOR): Water injection can induce clay swelling and pore throat reduction, while CO_2 injection may cause carbonate dissolution, alter wettability and reduce mechanical stability (Jin et al., 2017).
- Gas Hydrates: Thermal stimulation or depressurization causes dissociation of methane hydrates, producing internal gas and water, increasing pore pressure, and triggering strain localization or failure (Waite et al., 2009).
- Geothermal Energy: Continuous heat extraction causes thermal contraction, crack development, and permeability redistribution, all governed by strong THM couplings (Rutqvist, 2012).
- SAGD: Steam injection induces thermal expansion, pore pressure buildup, and caprock softening. The high temperatures accelerate material degradation and permeability evolution, requiring detailed modeling of both mechanical and thermal effects (Sheng, 2013).

Traditional simulators, such as those based solely on Biot's poroelasticity or Darcy flow, lack the capability to simulate localized damage, anisotropic transport, and feedback mechanisms involving:

- Hydration-induced swelling (e.g., smectite-rich shales),
- Chemical reactions (e.g., CO_2 –rock interactions),
- Fracture and fault reactivation,

- Permeability-percolation threshold effects due to throat collapse.

Such phenomena necessitate multi-scale, multi-physics frameworks that integrate both pore-scale and core-scale processes.

Advanced Experimental and Numerical Strategy

This study adopts a multi-scale THMC strategy that integrates high-resolution imaging, laboratory testing, finite element modeling, and artificial intelligence to advance understanding of coupled processes in porous media. The methodology includes:

- High-resolution X-ray micro-computed tomography (μ CT) and Digital Volume Correlation (DVC) to monitor time-lapse porosity, pore throat geometry, and fracture propagation in 3D (Cnudde & Boone, 2013; Zhang et al., 2022, Hamid et al., 2021, 2022, 2023, and 2024).
- Advanced triaxial and oedometer testing, coupled with ultrasonic and electrokinetic techniques, to quantify anisotropic stiffness, swelling coefficients, permeability evolution, and coupled electrochemical–mechanical responses (Revil et al., 2012; Jougnot et al., 2010).
- Fully coupled FEM simulations implementing hydro-chemo-poromechanical laws and damage mechanics, formulated within an extended Biot framework to model pore-scale and continuum-scale interactions (Lewis & Schrefler, 1998; Detournay & Cheng, 1993).
- Integration of Artificial Intelligence (AI)/ Machine Learning (ML) models, trained on simulation and experimental datasets, to provide real-time updates of permeability tensors, predict damage zones, and optimize flow redistribution forecasting. These models include XGBoost, CNNs, and LSTM networks for predictive modeling based on strain, temperature, damage, and cycle history.

AI/ML Integration in Coupled Porous Media Modeling

In recent years, the integration of AI and ML techniques into subsurface modeling has revolutionized the way coupled physical processes in porous media are understood, predicted, and managed. Traditional numerical approaches, such as Finite Element Models (FEM) and multiphysics solvers, remain essential for resolving the complex interactions between mechanical deformation, fluid transport, thermal effects, and chemical reactions. However, these methods are often computationally intensive and limited by their reliance on predefined constitutive relationships and idealized boundary conditions.

AI/ML frameworks offer a complementary paradigm by enabling data-driven learning from high-fidelity simulations, experimental datasets, and field measurements. When trained on carefully curated multiphysics data, machine learning models can identify nonlinear relationships, discover hidden patterns, and provide real-time predictions of system behavior under varying operational scenarios.

In the context of porous media, ML models such as XGBoost, Convolutional Neural Networks (CNNs), and Long Short-Term Memory (LSTM) networks have demonstrated high efficacy in capturing the evolution of key parameters including permeability, porosity, and damage under cyclic loading, thermal stress, and chemical exposure. These models are particularly valuable for forecasting performance in gas storage, EOR, and geothermal systems, where time-dependent changes in material properties critically impact long-term injectivity and productivity.

This study leverages AI/ML tools in conjunction with FEM simulations and experimental characterization to develop a hybrid predictive platform. The machine learning models are trained on simulation outputs and validated against experimental datasets derived from micro-CT imaging, triaxial testing, and permeability measurements. The framework enables real-time updates of permeability tensors, prediction of damage zones, and optimization of fluid flow redistribution in response to dynamic operational conditions.

By embedding AI/ML modules within a coupled hydro-mechanical-chemical simulation workflow, this research advances the state-of-the-art in digital rock physics and intelligent reservoir modeling. It not only

reduces computational costs but also enhances the ability to model heterogeneity, anisotropy, and path-dependent evolution—phenomena that are otherwise challenging to address through purely physics-based approaches.

This hybrid FEM–AI/ML platform enables dynamic updates of permeability, porosity, and mechanical damage across operational cycles and thermal/chemical changes, providing a powerful predictive tool for real-world subsurface applications.

Porous Media Characterization

Porous media are materials composed of a solid framework interspersed with a network of pores that can host fluids. These media are omnipresent in subsurface geomechanical systems, including petroleum reservoirs, aquifers, geothermal formations, and engineered storage facilities for hydrogen, CO₂, and natural gas. The behavior of porous materials under mechanical, hydraulic, thermal, and chemical influences governs critical phenomena such as permeability evolution, fracture initiation, and multiphase fluid migration.

Understanding the structure and dynamics of porous media is therefore fundamental to accurately modeling subsurface systems and predicting their evolution over operational timescales.

Structure of Porous Media

Porous media consist of two coupled continua (see Fig. 1):

- Solid Skeleton Continuum: Provides the mechanical support and deforms under stress.
- Fluid Continuum: Occupies the voids within the solid skeleton and is responsible for mass and energy transport.

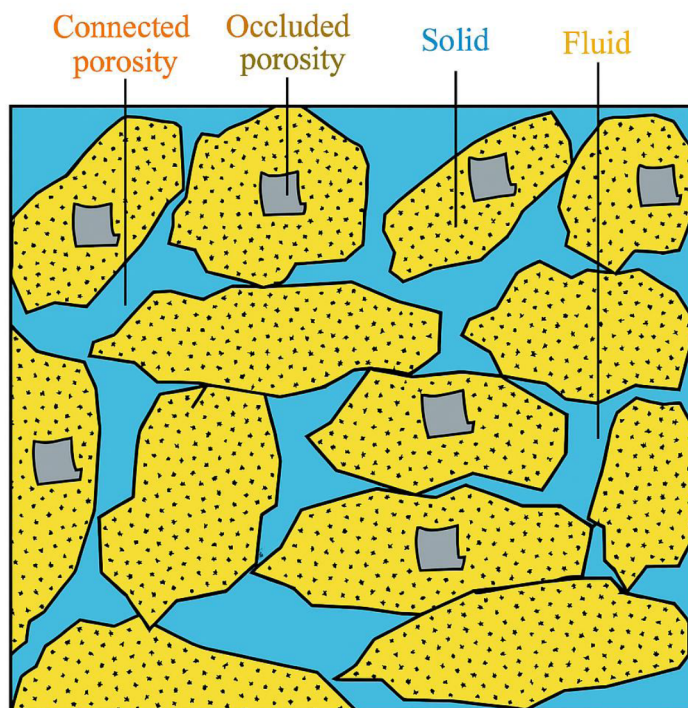


Figure 1—Porous media showing solid grains, connected porosity, occluded porosity, and fluid pathways (Modified after Zou 2020)

Solid Skeleton

The solid skeleton is formed by mineral grains whose arrangement and contact geometry define the mechanical behavior of the medium. The skeleton controls:

- Bulk elastic and plastic properties.
- Strength parameters such as cohesion and friction angle.
- Fracture initiation and propagation thresholds.

Fluid Continuum

The fluid continuum saturates the pore network, which includes:

- Connected Pores (Effective Porosity, ϕ_e): Facilitate flow.
- Isolated Pores (Occluded Porosity, ϕ_o): Do not contribute to flow directly.

The fluids may consist of:

- Single-phase systems: Water, oil, gas.
- Multiphase systems: Gas-water-oil coexistence, or gas-hydrate-brine systems.

Fluid properties (density, viscosity, interfacial tension) and pore morphology jointly govern the transport behavior.

Transport Phenomena and Pore Network Connectivity

Fluid flow and transport in porous media rely on pore connectivity, which enables continuous pathways for advection, diffusion, and dispersion. Poorly connected pores impede these processes, leading to low effective permeability even in highly porous media.

Coupled Pore Damage and Fluid Flow

Mechanistic Coupling

Porosity and permeability are inherently linked to pore throat geometry. Mechanical damage, chemical alteration, and thermally induced stresses cause:

- Reduction in pore throat radii (r_t) through microfracturing, grain rearrangement, or precipitation,
- Evolution of porosity (ϕ) through compaction or dilation,
- Permeability degradation (k) driven by reduction in connected pore network and throat sizes.

These effects dynamically feedback into fluid flow via modified permeability tensors, influencing local velocity fields, pressure gradients, and multiphase saturation distributions.

Mathematical Formulation

Permeability degradation due to evolving pore structure can be represented (Permeability update Kozeny-Carman + damage):

$$k(\epsilon, D, T, c) = k_0 \left(\frac{\phi}{\phi_0} \right)^n \exp(-aD)$$

Where:

- k_0 , ϕ_0 are initial permeability and porosity,

- D is a scalar damage variable (from 0 to 1),
- α is a degradation coefficient linked to throat collapse,
- n typically ranges from 2 to 4 (Kozeny-Carman-like).

Damage evolution coupling

$$D = D(\varepsilon_p, T, c, N)$$

Where:

D evolves with plastic strain ε_p , temperature T , chemical concentration c , and cyclic load history N .

Porosity reduction due to volumetric strain and damage

$$\phi = \phi_0 - (1 - \phi_0) \frac{\nabla V_s}{V_0} + f(D)$$

With volumetric strain computed from mechanical deformation:

$$\nabla V_s / V_0 = \text{tr}(\varepsilon_v)$$

$f(D)$ empirical function increasing porosity under microcracking.

$$\varepsilon_v = \nabla \cdot u$$

Flow Equation Update (Coupled Darcy's Law):

$$q = - \frac{k(\varepsilon, D)}{\mu} (\nabla p - \rho g)$$

Damage mechanics (Drucker-Prager plasticity with damage evolution):

$$F = a - p \tan \phi - c = 0$$

With $D = f(\varepsilon_p, T, N)$

ML Formulation (Example for Permeability Tensor k_{ij}):

$$k_{ij}^{ML} = f_{ML}(\varepsilon_p, T, S_w, N, D)$$

Where: f_{ML} is trained on FEM outputs and experimental data to capture complex, nonlinear couplings.

Pore throat radius reduction due to damage

Assuming the pore throat radius r_t reduces with cumulative damage $D \in [0, 1]$ and plastic strain history ε_p :

$$r_t = r_{t,0} (1 - \beta D)$$

Where:

$r_{t,0}$: Initial pore throat radius,

D : Damage parameter (from mechanical loading, swelling, thermal cycles),

β : Empirical reduction factor for throat damage ($0 < \beta \leq 1$)

Pore Size Reduction due to Volumetric Strain and Damage

Porosity evolves from volumetric strain and damage as:

$$\phi = \phi_0 - (1 - \phi_0) \text{tr}(\varepsilon_v) + \gamma D$$

Where:

ϕ_0 : Initial porosity,

$\text{tr}(\varepsilon_v) = \nabla \cdot u$: Volumetric strain,

γ : Porosity increase coefficient due to microcracks (typically $\gamma \ll 1$).

Alternatively, for compaction-dominated scenarios:

$$\phi = \phi_0(1 - D)$$

For dilation due to damage (crack opening)

$$\phi = \phi_0(1 + \gamma D)$$

Permeability Reduction Coupled with Pore Throat and Porosity Evolution

Kozeny–Carman Equation with Damage incorporating pore throat collapse ($r_t \downarrow$)

$$k = k_0 \left(\frac{r_t}{r_{t,0}} \right)^2 = k_0 (1 - \beta D)^2$$

Final General Formulation Combining Porosity, Pore Throat, and Damage

$$k = k_0 \left(\frac{\phi}{\phi_0} \right)^n (1 - \beta D)^2 e^{-\alpha D}$$

Where:

- α : Additional exponential degradation factor capturing microcracking effects,
- $(1 - \beta D)^2$: Pore throat reduction contribution,
- $(\phi/\phi_0)^n$: Porosity effect,
- $e^{-\alpha D}$: Additional permeability degradation beyond geometry (anisotropic damage pathways).

Cumulative Damage Evolution Law (General Form)

Damage accumulation due to plastic strain ε_p , temperature T , and cycle count N :

$$D = 1 - \exp(-\delta(\varepsilon_p + \lambda T + \zeta N))$$

Where:

- δ, λ, ζ : Empirical constants,
- ε_p : Equivalent plastic strain,
- T : Temperature,
- N : Load/unload cycles.

Methodology

Experimental Methodology

A comprehensive experimental program was designed to investigate the coupled mechanical and transport behavior of porous media under representative subsurface conditions relevant to gas production, storage, and cyclic operations.

Micro-CT Imaging and Digital Rock Physics

High-resolution micro-computed tomography (μ CT) was utilized to capture detailed three-dimensional images of pore structures in clastic core plugs before and after mechanical loading cycles. Digital Volume Correlation (DVC) techniques were employed to monitor strain localization and deformation at the microstructural scale. Pore and throat size distributions, connectivity, and porosity metrics were extracted to evaluate the impact of mechanical loading on flow pathways and overall pore network integrity.

Mechanical and Transport Testing

Triaxial compression tests were conducted on core plug samples to replicate reservoir-like stress conditions and to measure deformation behavior under cyclic loading. Acoustic wave velocities (P-wave and S-wave) were recorded in real-time to assess elastic property evolution and to detect damage progression. Permeability measurements were performed using steady-state flow apparatuses before and after each loading stage to quantify changes in fluid transport capacity due to mechanical deformation and damage accumulation.

Electrokinetic and Fluid Flow Characterization

Electrokinetic tests, including electrical conductivity and streaming potential measurements, were conducted to analyze ion transport behaviors under stress. Complementary fluid flow experiments provided insights into the dynamic changes in permeability, connectivity, and flow pathways as a function of progressive mechanical damage and pore structure evolution.

Numerical Modeling Approaches

A fully coupled finite element modeling (FEM) framework was developed to simulate the interactions between mechanical deformation, damage accumulation, and permeability evolution within porous media. The model integrated hydro-mechanical processes and accounted for feedback mechanisms between pore structure degradation and fluid flow. Domain geometries were constructed from μ CT-derived pore structures, ensuring realistic representation of complex pore networks.

The simulations were designed to replicate experimental conditions, incorporating cyclic loading scenarios to capture time-dependent variations in mechanical and transport properties. Outputs included temporal evolution of porosity, permeability, fluid velocity distributions, and localized damage zones.

Machine Learning Integration

Machine learning models were developed and trained using datasets from experimental measurements and FEM simulations. These models provided real-time predictions of permeability evolution and damage progression based on input parameters such as mechanical strain, temperature history, saturation, and cyclic loading counts. Algorithms employed included XGBoost, convolutional neural networks (CNNs), and long short-term memory (LSTM) networks, selected for their ability to capture complex, nonlinear relationships between evolving material properties and flow behaviors.

Validation and Data Analysis

Simulation results were validated against experimental data obtained from μ CT imaging, permeability testing, and mechanical measurements. Post-processing focused on analyzing changes in pore connectivity, permeability distributions, and damage localization patterns. Visualization tools were used to illustrate pore structure evolution, flow velocities, and anisotropic damage propagation, providing a robust interpretation of the coupled processes governing porous media behavior under operational loading conditions.

Results

Understanding the evolution of pore structure under mechanical loading is critical for accurately modeling flow and transport in porous media. Digital Rock Physics (DRP), combined with micro-CT imaging, provides quantitative insights into pore throat connectivity, pore size distribution, and the irreversible changes resulting from mechanical cycling.

Experimental Characterization And Image-Based Modeling Of Porous Media Using Digital Rock Physics Microstructural Analysis Pre- and Post-Loading

Fig. 2: Presents high-resolution micro-CT images of a representative porous rock sample before and after mechanical loading cycles. Subfigure (a) illustrates the intact microstructure, characterized by well-connected pore networks and clearly defined throats. Subfigure (b) demonstrates substantial alterations post-loading, including throat constriction, pore body shrinkage, and connectivity degradation. Scale bars range from 1 mm to 100 μ m, highlighting microstructural evolution across scales.

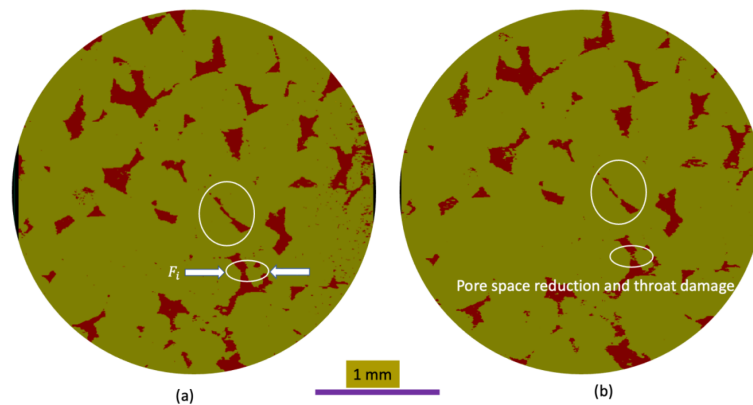


Figure 2—Micro-CT cross-sections of a porous rock sample: (a) Initial structure with visible pore throats and pore bodies, with directional force vectors (F_t) indicated. (b) Post-loading structure exhibiting pore throat closure and pore space reduction. Structural changes are irreversible due to mechanical damage and compaction effects (After Hamid et al. 2021, 2023).

Pore Throat Closure and Damage Localization

Prior to loading (Figure 2a), discrete throats facilitate fluid connectivity through a continuous pore network. Post-loading (Figure 2b) reveals closure of these conduits, predominantly due to compaction and localized shear stresses. This structural degradation aligns with established mechanisms of permeability reduction in geomechanics, wherein mechanical deformation concentrates at constriction points.

Pore Volume Reduction and Microstructural Collapse

Comparative analysis indicates a reduction in isolated pore bodies' size, a consequence of inelastic deformation processes such as grain rearrangement, plastic collapse, and compaction-induced porosity loss. These effects are especially pronounced in rocks containing clay or other ductile mineral phases susceptible to deformation under repeated loading cycles.

Irreversibility and Structural Memory

Post-deformation structures exhibit persistent alterations, with neither pores nor throats reverting to their initial geometries. This indicates irreversible, damage-driven deformation consistent with cumulative mechanical fatigue, leading to permanent loss in porosity and permeability.

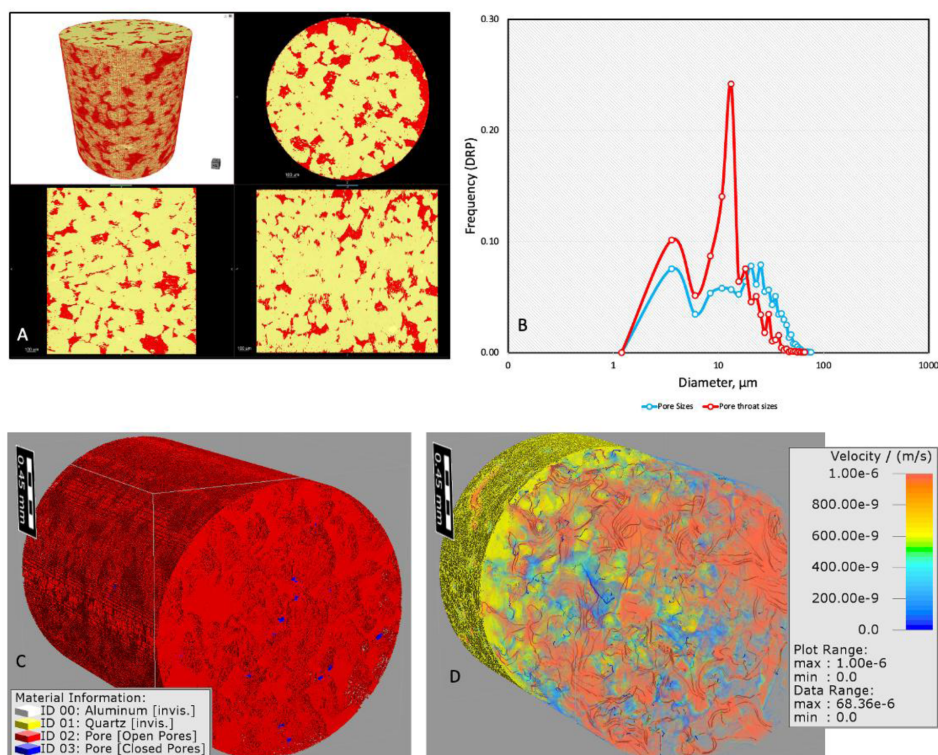
Force Pathway and Strain Localization

Tangential force (F_t) directions, depicted in Figure 2a, induce localized shear stresses concentrated at pore throats. Due to high surface-area-to-volume ratios, these regions are especially vulnerable to closure, a finding consistent with known behaviors of throat regions under deviatoric loading regimes.

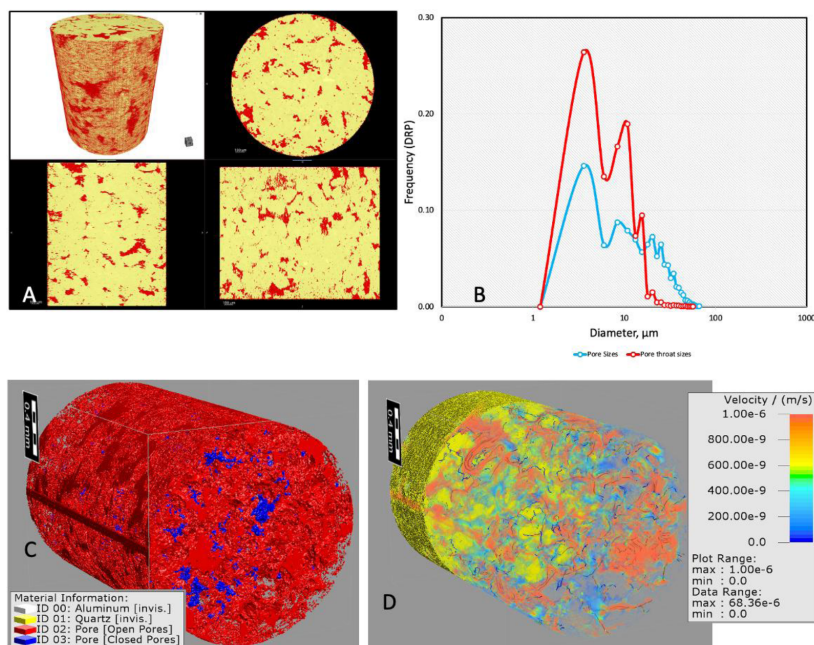
Implications for Reservoir Permeability and Flow Functionality

- **Hydraulic Connectivity:** Throat closure directly reduces hydraulic pathways, degrading absolute and relative permeability.
- **Capillary Pressure:** Reduced connectivity modifies capillary pressure curves and phase displacement behavior.
- **Flow Dynamics:** Diminished pore network connectivity affects breakthrough times, relative permeability ratios, and overall fluid mobility.
- **Cyclic Loading:** In scenarios such as underground gas storage or cyclic injection, these mechanisms drive permeability hysteresis and cumulative flow degradation.

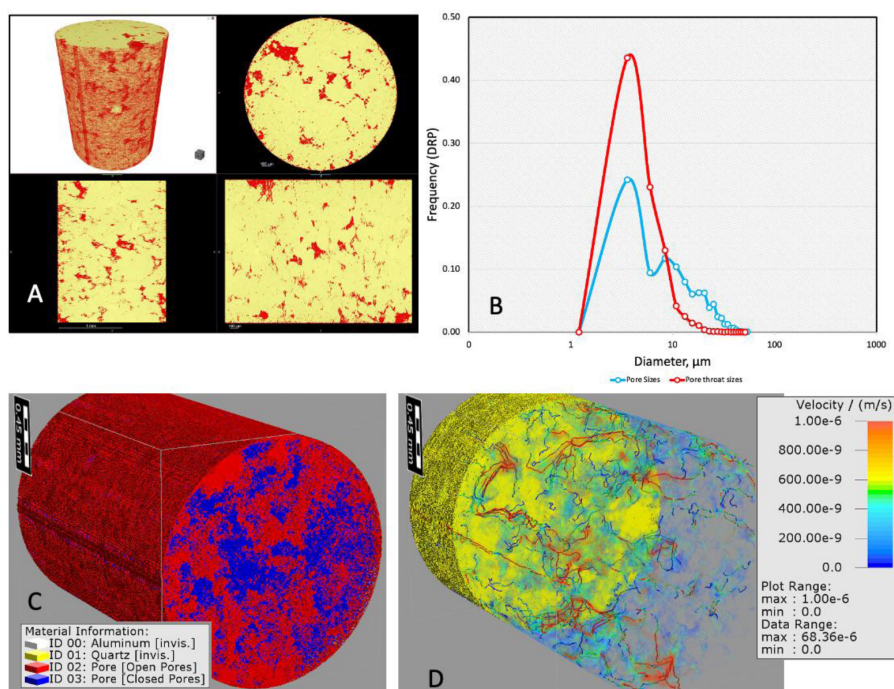
Fig 3 to 6 illustrate the evolution of pore structure, connectivity, and permeability in clastic core plug samples subjected to progressive mechanical loading and unloading cycles. This dataset captures the transition from high-permeability, well-connected porous systems to low-permeability, degraded microstructures, providing a clear visualization of pore-throat damage and its direct impact on fluid flow behavior.



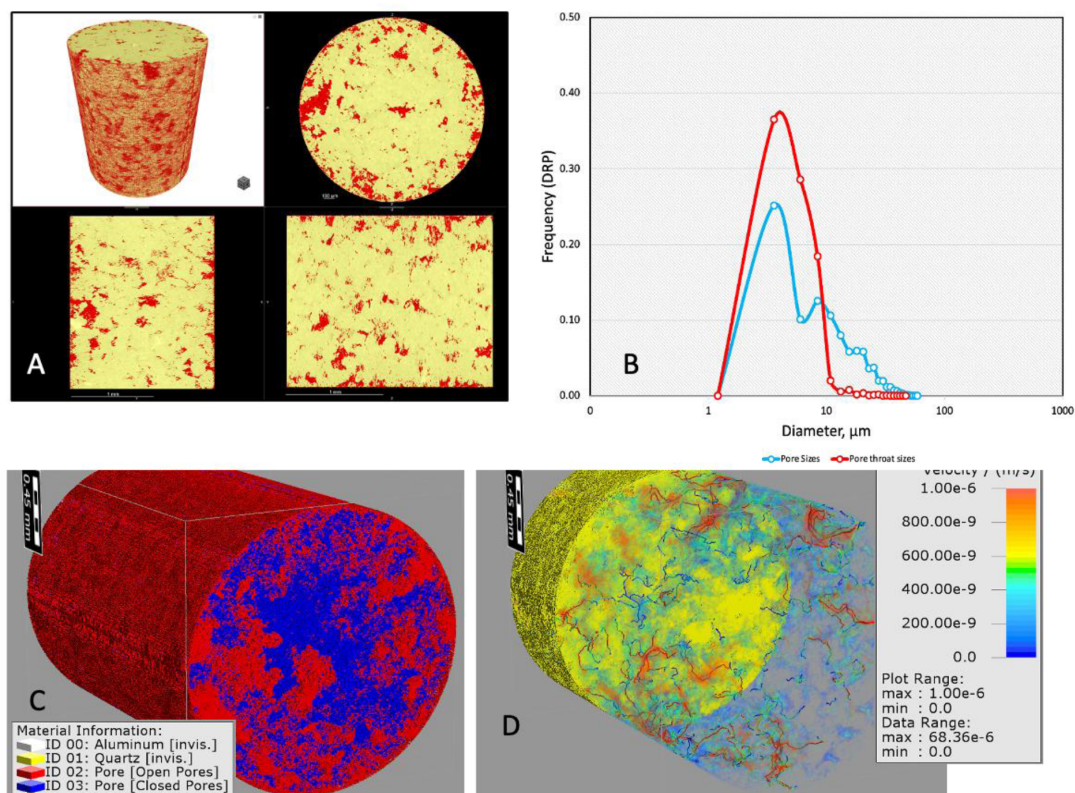
Figures 3—Progressive evolution of pore structure, throat connectivity, and flow behavior in clastic core plugs under cyclic mechanical loading. (A) 3D micro-CT images show pore space degradation; (B) Pore and throat size distributions shift toward smaller sizes with damage; (C) high porosity, high pore connectivity (D) Flow simulations visualize the transition for high-permeability continuous pathways.



Figures 4—Progressive evolution of pore structure, throat connectivity, and flow behavior in clastic core plugs under cyclic mechanical loading. (A) 3D micro-CT images show pore space degradation; (B) Pore and throat size distributions shift toward smaller sizes with damage; (C) high to medium porosity, with some closed pore space (D) Flow simulations visualize the transition for high to medium-permeability continuous pathways.



Figures 5—Progressive evolution of pore structure, throat connectivity, and flow behavior in clastic core plugs under cyclic mechanical loading. (A) 3D micro-CT images show pore space degradation; (B) Pore and throat size distributions shift toward smaller sizes with damage; (C) medium to low-porosity, with abundance of closed pore space; (D) Flow simulations visualize the transition for medium to low-permeability continuous pathways.



Figures 6—Progressive evolution of pore structure, throat connectivity, and flow behavior in clastic core plugs under cyclic mechanical loading. (A) 3D micro-CT images show pore space degradation; (B) Pore and throat size distributions shift toward smaller sizes with damage; (C) low porosity, with increasing closed pore space; (D) Flow simulations visualize the transition for low-permeability continuous pathways.

Fig. 3A: Shows a 3D Micro-CT visualization of pore structure (Pre-Damage). A well-connected, highly porous structure is evident. Red regions denote pores within a continuous solid matrix (yellow). Multiple orthogonal slices show uniform pore connectivity. The open and continuous pore network in all directions explains the high intrinsic permeability. There is minimal evidence of pore isolation, indicating good percolation across the sample.

Fig. 3B: Shows pore and throat size distributions, Bimodal distributions for both pore sizes (blue) and pore throat sizes (red) are observed. Pore sizes span from $\sim 1 \mu\text{m}$ to $>100 \mu\text{m}$. Throat sizes peak below $10 \mu\text{m}$ but extend into larger ranges. The alignment of throat and pore distributions confirms good connectivity with no severe throat bottlenecks.

Fig. 3C: Depicts pore connectivity classification, open (red) vs. closed Pores (blue). Segmentation distinguishes between. Open pores dominate the sample, confirming that most porosity contributes directly to permeability, while closed pores represent a negligible fraction of the volume. High effective porosity supports robust permeability, ideal for applications like waterflooding, CO_2 storage, or primary recovery.

Fig. 3D: Represents velocity field simulation (Flow Characteristics). Simulated flow fields show velocities up to 10^{-6} m/s through the pore network. Streamlines indicate preferential flow aligned with the applied pressure gradient.

Fig 4A: 3D Micro-CT visualization of pore structure (Post-Damage). The core plug still contains a connected pore network (red), but compared to the previous high-permeability case, there is a noticeable reduction in pore volume and throat connectivity. Mechanical loading has caused compaction and partial closure of pores and throats, especially in isolated regions. Connectivity in the axial direction persists but is less robust laterally.

Fig 4B: Pore and throat size distribution (Post-Damage). The throat size distribution (red) shows a strong reduction in larger throats compared to the previous high-permeability case. The dominant throat sizes have shifted towards the smaller end (sub- $10 \mu\text{m}$). Pore sizes (blue) still exhibit a broad distribution but are narrower than before. Mechanical damage has disproportionately affected larger throats through closure and compaction.

Fig 4C - Pore connectivity classification (Open vs. Closed Pores). There is an increase in closed pores (blue) relative to the prior case, with open pores (red) still connected but less uniformly distributed. Progressive cycles of loading/unloading have isolated some pore spaces. Open pore connectivity remains but is constrained to fewer pathways. The increase in closed pores correlates with decreased effective porosity and permeability.

Fig 4D - Velocity field simulation (Flow Characteristics Post-Damage). Flow velocities are lower and more spatially restricted than in the previous high-permeability sample. Velocity gradients suggest tortuous, less efficient pathways. Maximum velocities remain around 10^{-6} m/s but occupy reduced volumes. Flow is increasingly localized into preferential pathways, while tortuosity and flow path lengthening indicate reduced permeability. The reduction in flow efficiency aligns with a medium-permeability classification. Flow anisotropy may increase, and effective permeability may decrease significantly under continued cycling.

Fig. 5A: 3D Micro-CT visualization of pore structure (Significant Damage). The pore network (red) shows pronounced disconnection and reduced pore volume compared to previous medium-permeability cases. Pores appear fragmented and isolated. Mechanical loading and cumulative compaction have closed many smaller throats and disconnected portions of the pore space, severely compromising connectivity.

Fig. 5B: Pore and throat size distribution (Post-Damage). Both pore and throat size distributions have shifted further towards smaller diameters compared to previous samples:

- Throat sizes (red) peak sharply at small diameters ($<5 \mu\text{m}$).
- Pore sizes (blue) exhibit a narrower, lower-frequency distribution.

- The reduction in larger throats suggests widespread compaction and mechanical closure.
- Smaller throats dominate, which increases capillary pressures and restricts fluid flow.

Fig. 5C: Pore connectivity classification (Increased Closed Pores). A substantial increase in closed pores (blue) relative to open pores (red) is evident, with open pores now localized and less interconnected.

- Damage mechanisms have significantly isolated pores.
- Remaining open pores form smaller, less connected clusters.
- Effective porosity is reduced sharply; many regions no longer contribute to flow.

Fig. 5D: Velocity field simulation (Degraded Flow Characteristics). The velocity field is highly localized with reduced flow regions. High-velocity zones are sparse and constrained to narrow channels, indicating tortuous pathways. Maximum velocities are lower in extent and distribution.

- Fluid flow is increasingly constrained to preferential, tortuous paths.
- Tortuosity and flow heterogeneity dominate, reducing overall permeability.
- Flow is inefficient, aligned with medium-to-low permeability rocks. Significant regions act as stagnant zones with minimal contribution to flow.

Fig. 6A: 3D Micro-CT visualization of pore structure (Severely Damaged). Very sparse and disconnected pore networks remain. Pores are small, isolated, and poorly connected across the rock volume. Cumulative mechanical loading has effectively collapsed most pore throats and pinched off formerly connected pore spaces. The rock now behaves like a tight formation.

Fig. 6B: Pore and Throat Size Distribution (Collapse of Connectivity). Both pore size (blue) and throat size (red) distributions are narrow and shifted toward the smallest diameters ($<5\ \mu\text{m}$). Throat sizes dominate at sub-micron to a few microns, with almost no larger throats present. Severe mechanical closure of throats and pores has resulted in uniform, minimal connectivity features.

Fig. 6C: Pore connectivity classification (Predominantly Closed Pores). Closed pores (blue) dominate the sample, with open pores (red) confined to small, isolated patches. Connectivity between open pores is extremely limited.

- The sample has transitioned from a connected porous medium to a partially occluded structure with negligible hydraulic continuity.
- Effective porosity is extremely low; many pores are occluded and do not contribute to flow. This severely limits permeability.

Fig. 6D: Velocity field simulation (Extremely Restricted Flow). The velocity field shows minimal flow with highly localized, sporadic high-velocity channels through rare, connected throats. Overall, flow velocity is low and discontinuous.

- Flow is dominated by isolated, tortuous micro-channels.
- Large regions exhibit near-stagnant behavior.
- Bulk permeability is extremely low, likely $<1\ \text{mD}$, with significant flow anisotropy and negligible contribution from most of the volume.

Implications for Reservoir Management

- Progressive damage from cyclic loading drives irreversible reductions in permeability.

- Flow transitions from uniform and isotropic to highly localized and anisotropic.
- Effective porosity shrinks as pore isolation increases with throat closure.
- Capillary behavior shifts, impacting recovery strategies and storage operations.
- Reservoir models must account for cumulative damage, especially in cyclic scenarios (gas storage, EOR).

This study demonstrates the utility of micro-CT-based digital rock physics for capturing the coupled mechanical and hydraulic evolution of porous media under cyclic loading. Key findings include:

- Irreversible reduction in pore throat dimensions leads to permanent permeability loss.
- Pore size shrinkage and throat closure are primary mechanisms for porosity and connectivity reduction.
- Cyclic loading amplifies damage, exacerbating flow pathway degradation and inducing hysteresis.
- In high-permeability cases, well-connected pore networks enable efficient fluid transport with minimal resistance.
- Repeated loading cycles compound microstructural damage, reinforcing the necessity to account for time-dependent degradation in reservoir models.

These results have direct implications for modeling subsurface operations, particularly where cyclic loading governs reservoir performance

Numerical Simulation Results

Numerical simulation is a crucial tool in the study of porous media, particularly for predicting the evolution of microstructural properties (such as pore throat size, pore size, porosity, and permeability) under various physical mechanisms, including mechanical damage, chemical interactions, and fluid flow. Understanding how damage propagates within porous rocks directly affects the ability to predict reservoir behavior, production performance, and long-term integrity under cyclic or continuous loading conditions.

Fig. 7: Illustrates the progressive degradation of a porous medium subjected to a damage-driven process over time, such as mechanical compaction, chemical reactions, or both.

Fig. 7A: Shows pore throat radius reduction over time, the pore throat radius decreases almost linearly from 1.0 microns to approximately 0.9986 microns over the simulation period. This reduction indicates progressive narrowing of pore throats, likely due to mechanical compaction or damage-driven closure mechanisms.

Fig. 7B: Depicted pore size reduction over time, the main pore body size decreases from 10 microns to approximately 9.986 microns. Pore body reduction follows a similar trend to the throat reduction, but the magnitude of change is proportionally smaller compared to throat reduction. This reflects the inherent mechanical hierarchy: throats close faster under stress, whereas larger pores resist deformation more.

Fig. 7C: Porosity reduction over time (Damage-Driven)

Observation: Porosity reduces from 0.20 to approximately 0.19984 over time, this small but consistent decline reflects a densification of the porous structure due to damage accumulation or mechanical compaction, which progressively reduces the total void space available for fluids.

Fig. 7D: Permeability Reduction Over Time (Damage-Driven), permeability decreases from approximately $9.86 \times 10^{-16} \text{ m}^2$ to around $9.74 \times 10^{-16} \text{ m}^2$. Permeability reduction mirrors the geometric reductions in pore and throat sizes and porosity. Since permeability is highly sensitive to throat radii, even minor reductions yield significant permeability loss. This decline will reduce the efficiency of fluid

extraction/injection processes, adversely affecting production forecasts in reservoir engineering or storage capacity in CO₂ or H₂ sequestration.

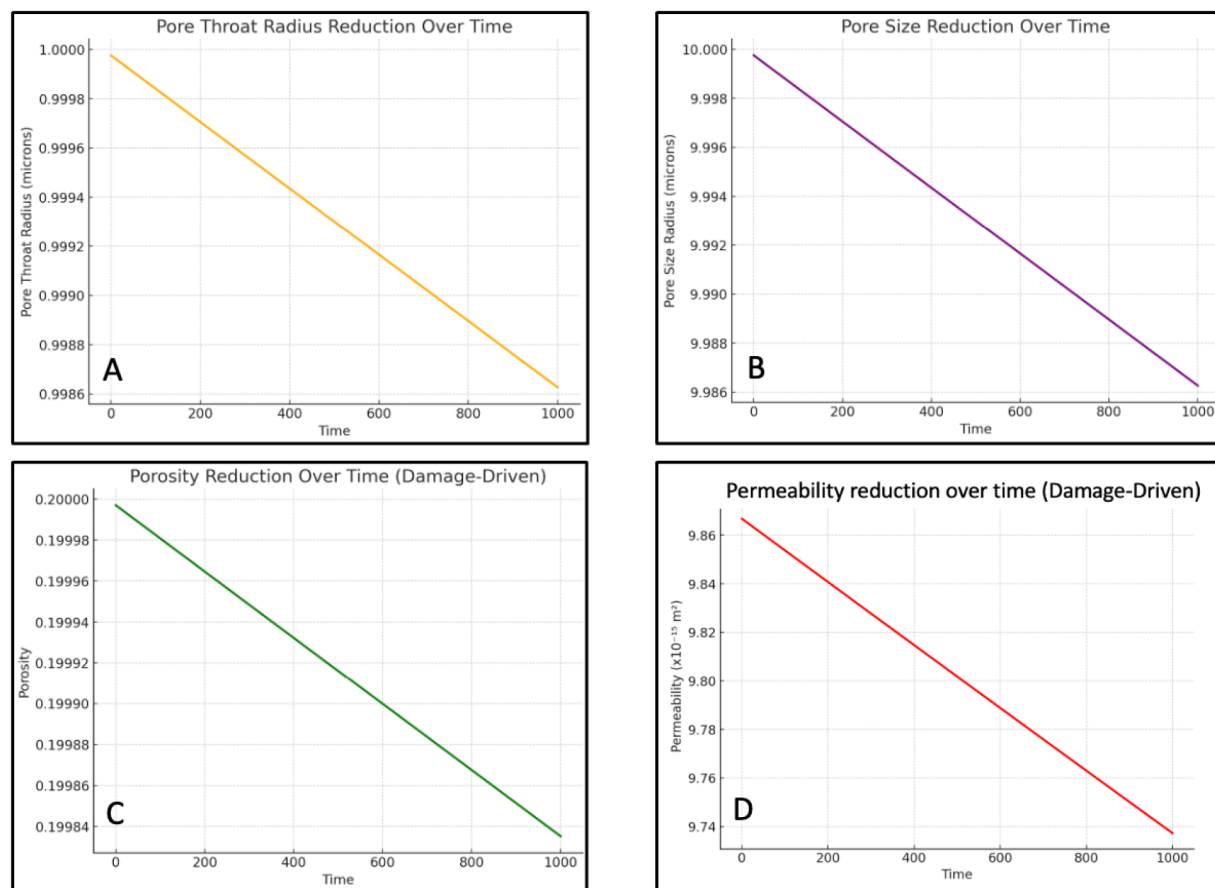


Figure 7—Numerical simulation results illustrating the progressive degradation of porous media properties under damage-driven mechanisms over time

Coupled Evolution of Pressure, Damage, and Permeability

The numerical results presented in Fig. 8 demonstrate the complex evolution of pressure distribution, pore throat damage, and permeability within a porous medium subjected to a differential pressure gradient, representative of subsurface processes such as cyclic gas injection, enhanced oil recovery (EOR), or geothermal operations. The simulation incorporates fully coupled hydromechanical interactions, capturing the time-dependent feedback mechanisms between mechanical deformation and fluid transport properties.

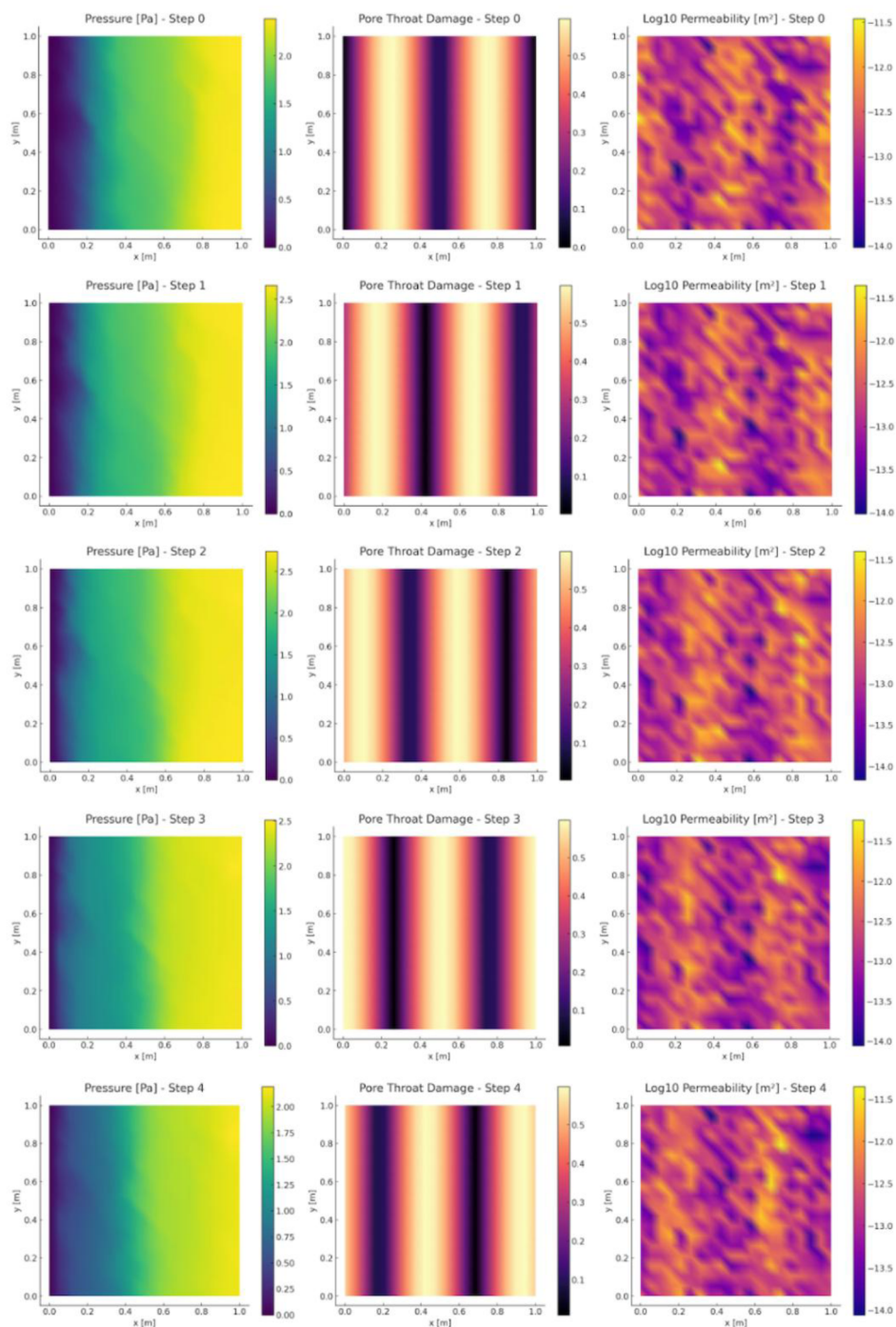


Figure 8—Time-step multiphysics simulation of coupled fluid flow, mechanical damage, and permeability evolution in a sandstone formation.

Pressure Distribution Evolution

The imposed boundary conditions establish a stable left-to-right pressure gradient across the domain, simulating injection or production scenarios. Over successive timesteps, minor modifications in the pressure field geometry become evident, reflecting the influence of evolving permeability and damage on flow pathways. Specifically, a progressive flattening of pressure contours indicates the emergence of preferential flow channels in regions of increased permeability, a behavior consistent with field observations in fractured or stimulated reservoirs where heterogeneities govern fluid redistribution.

Pore Throat Damage Propagation

Pore throat damage initiates in discrete vertical bands, suggesting the presence of pre-existing weaknesses or mechanically compliant zones within the microstructure. As the simulation progresses, these damaged zones expand laterally and vertically, driven by pore pressure escalation and resultant effective stress reduction. The damage parameter employed in the model captures the cumulative effects of microstructural degradation mechanisms, including crack propagation, grain boundary sliding, and shear dilation, all of which contribute to the modification of local transport properties. The observed anisotropy in damage evolution underscores the necessity of considering directionally dependent material behavior in predictive models.

Permeability Enhancement and Nonlinear Response

Permeability fields, presented on a logarithmic scale, reveal a distinct transition from initial low-permeability conditions ($k \approx 10^{-14} m^2$) characteristic of tight formations, to enhanced permeability zones reaching ($k \approx 10^{-11.5} m^2$) in localized regions by the final timestep. The spatial correlation between regions of moderate damage and significant permeability increase highlights a critical aspect of the damage-permeability relationship: while moderate damage can improve connectivity and facilitate flow, excessive degradation may lead to compaction, pore closure, or localized permeability reduction. This nonlinear behavior is reflective of complex subsurface responses where pore-scale phenomena aggregate to influence macroscopic flow behavior.

Implications for Reservoir Performance and Modeling

These findings emphasize the strongly coupled nature of mechanical deformation and transport properties within porous media under operational loading conditions. The simulation results demonstrate that relatively minor mechanical changes at the pore throat scale can precipitate substantial alterations in flow pathways and permeability tensors. This underscores the importance of incorporating dynamic permeability updates within reservoir simulators, particularly for applications involving cyclic stress regimes.

Furthermore, the hysteretic nature of permeability evolution observed herein aligns with experimental and field data indicating that damage-induced permeability changes are not readily reversible. This has significant implications for the long-term management of subsurface assets, where cyclic operations may lead to progressive, cumulative degradation of reservoir quality. Recognition and quantification of these effects are essential for accurate forecasting of reservoir performance, optimization of injection and production strategies, and ensuring the geomechanical integrity of storage and production intervals.

This framework (Fig.9) is a cutting-edge integration of Artificial Intelligence (AI) and Finite Element Method (FEM) for modeling and forecasting damage, permeability, and porosity evolution in porous media under coupled thermo-hydro-mechanical-chemical (THMC) conditions. It is designed to support real-time, high-fidelity simulation and prediction in subsurface systems such as hydrogen storage, CO₂ sequestration, unconventional reservoirs, and geothermal energy.

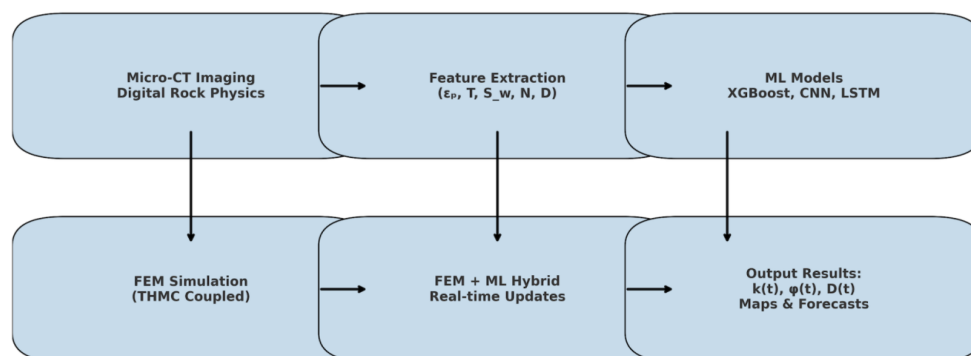


Figure 9—Shows evolution of mechanical and transport properties over time reveals intricate cyclic responses of porous media under coupled Thermo-Hydro-Mechanical-Chemical (THMC) loading. This behavior is especially relevant for hydrogen storage, cyclic gas injection, geothermal systems, and swelling/shrinking cement or rocks.

Fig. 10: Illustrates the temporal evolution of key geomechanical and transport properties over a 30-day period in a porous medium under coupled THMC (Thermo-Hydro-Mechanical-Chemical) processes:

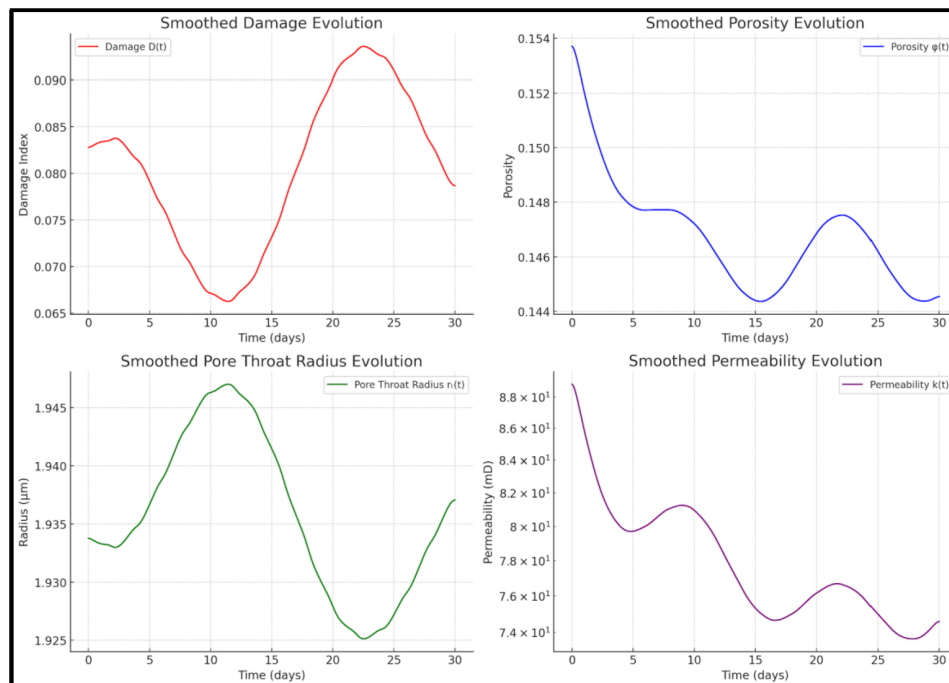


Figure 10—Smoothed evolution of damage index $D(t)$, porosity $\phi(t)$ pore throat radius ($r_t(t)$), and permeability $k(t)$ time under cyclic THMC loading.

Damage Evolution $D(t)$:

The cyclic rise and fall in the damage index indicates alternating phases of mechanical degradation (microcrack propagation due to pressure or thermal spikes) and recovery or stabilization (e.g., stress unloading, mineral healing, thermal relaxation).

Such behavior mimics field-representative cyclic operations (e.g., injection/withdrawal, seasonal temperature variation).

Porosity Evolution $\phi(t)$:

The porosity declines during compaction, plastic strain, and fine migration.

Intermittent increases arise from damage-induced dilation, microcrack opening, or unloading episodes.

These trends underscore the competition between compaction and dilation, a critical mechanism in time-dependent geomechanical systems.

Pore Throat Radius $r_t(t)$:

The pore throat radius reflects similar cyclic variations, decreasing under compaction and plugging, and increasing during dilation or thermal expansion.

This directly controls permeability evolution, highlighting its sensitivity to microscopic structure fluctuations.

Permeability Evolution $k(t)$:

The permeability exhibits non-monotonic trends, initially decreasing with compaction, then partially recovering due to dilation or damage re-opening.

These dynamics capture realistic permeability hysteresis, common in tight gas reservoirs, cement systems, and shale formations subjected to thermal or mechanical cycling.

The AI/ML-coupled model integrates microstructural and operational features to simulate sinusoidal trends associated with thermal stimulation, pressure cycling, and material degradation-recovery phases. Permeability and porosity exhibit non-linear responses due to the interaction between compaction, dilation,

and evolving damage, highlighting the dynamic behavior of porous media under cyclic stress and temperature conditions.

Conclusion

This study provides a comprehensive investigation into the coupled mechanical, hydraulic, and microstructural evolution of porous media subjected to cyclic loading conditions, representative of subsurface gas production, storage, and energy recovery operations. The integration of high-resolution micro-CT imaging, advanced laboratory testing, digital rock physics, finite element modeling, and machine learning offers a robust framework for analyzing pore-scale damage mechanisms and their impact on macroscopic flow properties.

The experimental results reveal a clear progression of pore throat closure, porosity reduction, and connectivity degradation with increasing mechanical loading cycles. These changes lead to irreversible reductions in permeability, increased flow tortuosity, and anisotropic transport behaviors. Numerical simulations successfully captured the observed phenomena, demonstrating the strong coupling between mechanical damage and permeability evolution. Machine learning models enhanced the predictive capability of permeability updates, providing a valuable tool for real-time assessments in field-scale simulations.

- Pore throat damage is the dominant mechanism driving permeability reduction under cyclic loading.
- Damage accumulation leads to significant flow path restriction and pore isolation, transitioning systems from high to low permeability states.
- Coupled modeling approaches are essential to accurately capture the dynamic interplay between mechanical and hydraulic processes.
- Machine learning integration improves the efficiency and adaptability of predictive models for field applications.

These insights are critical for improving the understanding of reservoir behavior under cyclic operations and enhancing the design of strategies for underground gas storage, enhanced oil recovery, and geothermal energy production.

Recommendations

Based on the findings of this study, the following recommendations are proposed for both research and field applications

- Incorporate Coupled Modeling in Field Simulations: Reservoir models should integrate coupled hydro-mechanical formulations with evolving permeability and damage to improve the accuracy of long-term performance predictions under cyclic loading conditions.
- Utilize High-Resolution Imaging for Critical Reservoirs: For reservoirs subjected to repeated injection and production cycles, high-resolution μ CT imaging and digital rock physics should be employed to characterize initial pore structure and identify zones susceptible to damage.
- Adopt Machine Learning for Real-Time Monitoring: Machine learning frameworks, trained on experimental and numerical datasets, offer efficient tools for real-time updating of permeability and damage parameters in reservoir simulations, enhancing forecasting reliability.
- Account for Irreversible Changes in Operational Planning: Field operations, particularly for gas storage and cyclic EOR, should consider the irreversible nature of permeability and porosity degradation due to mechanical damage when designing injection and withdrawal schemes.

- Future Research Directions: Further studies are recommended to extend this framework to include thermo-chemo-mechanical coupling, multiphase flow scenarios, and fault-fracture interactions, providing a more holistic understanding of subsurface behavior.

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